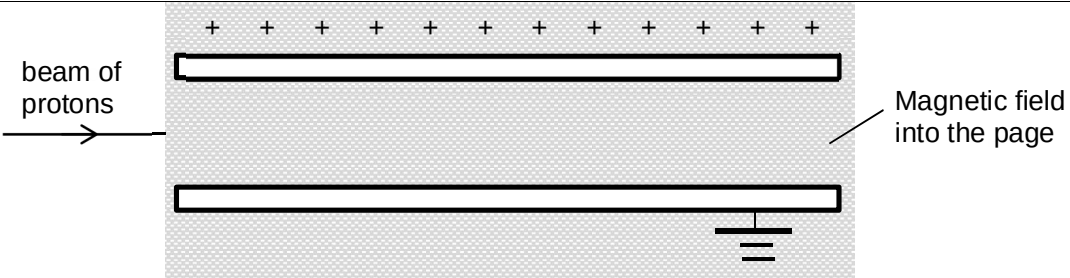


22

A velocity selector for protons has an electric field of strength $3.0 \times 10^5 \text{ V m}^{-1}$, produced by two horizontal plates. The upper plate is connected to a positive potential and the lower plate is earthed. A magnetic field of flux density $1.5 \times 10^{-2} \text{ T}$, directed into the plane of the paper, is at right-angles to the electric field as shown in the diagram.



beam of protons

Magnetic field into the page

Which of the following could be the possible speed of a proton and its observation?

		speed	observation
	A	$1.8 \times 10^7 \text{ m s}^{-1}$	undeflected
	B	$2.0 \times 10^7 \text{ m s}^{-1}$	deflected downwards
	C	$2.2 \times 10^7 \text{ m s}^{-1}$	deflected downwards
	D	$2.4 \times 10^7 \text{ m s}^{-1}$	deflected upwards

Answer: D

Protons entering the velocity have a range of speeds.

If protons are undeflected, Upwards magnetic force = Downwards electric force.

i.e. $Bqv = qE$

$$v = E/B = 3.0 \times 10^5 / 1.5 \times 10^{-2}$$

$$= 2.0 \times 10^7 \text{ m s}^{-1}.$$

Protons with speeds higher than this value will have a greater magnetic force, but the electric force is constant regardless of speed, hence deflected upwards.

Conversely, protons with speeds lower than this value will deflect downwards.